



BookIt

Appointment Booking System

Part 1 of 2 — **Product & Features Walkthrough**

Book anything — doctors, salons & turfs, with zero double-bookings.

React

Vite

TypeScript

Express

PostgreSQL

Zod

JWT

Prepared 14 July 2026 · Author: Bhanu Prakash

o • What BookIt is

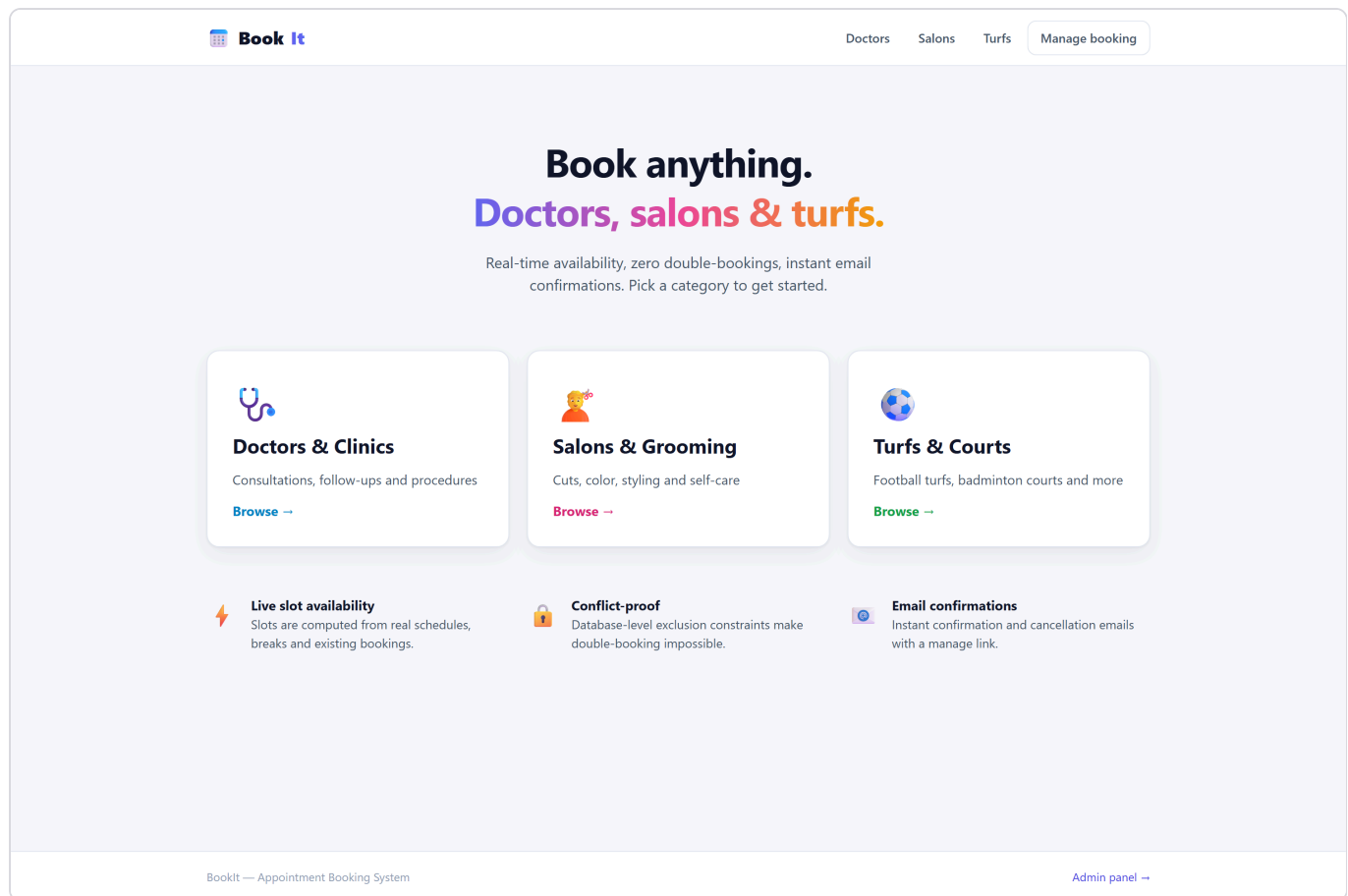
A full-stack, multi-vertical appointment booking platform for three business types — **doctors, salons, and sports turfs/courts**. Customers browse providers, see real-time bookable slots computed from each provider's live schedule, and book or cancel — while the backend makes **double-booking impossible**. Admins manage providers, schedules, services, and bookings.

The interviewer one-liner: "It's a Vite + React SPA talking to an Express + PostgreSQL API. The engineering centerpiece is that a fixed slot can never be sold twice under concurrency — I enforce it with three layers: a per-provider advisory lock, in-transaction re-validation, and a Postgres exclusion constraint that turns any overlap into an HTTP 409."

Tech stack

LAYER	CHOICE
Frontend	React 18 + react-router 6, TypeScript, Vite 6 (SPA, plain CSS)
Backend	Node.js (ESM) + Express 4, TypeScript via <code>tsx</code>
Database	PostgreSQL 13+ with raw SQL via <code>pg</code> (no ORM); <code>btree_gist</code> , GiST exclusion constraints, a custom <code>timerange</code> type
Validation	Zod on every request body/query
Auth	Admin-only JWT (12h) + bcrypt; customers identified by booking code + email (no account)
Email	nodemailer (dev writes rendered HTML to <code>server/outbox/</code>)

1 • The customer site



Home — three category cards (doctors, salons, turfs) + the feature strip

- **Home** — a hero and three category cards, plus a feature strip ("Live slot availability", "Conflict-proof", "Email confirmations").
- **Browse by category** (`/browse/:type`) — the provider list for doctors / salons / turfs, with search.

Doctors & Clinics

Search by name or specialty...

**Dr. Asha Rao**
Cardiologist

15+ years in preventive cardiology. MBBS, MD (Cardiology). Known for unhurried consultations.

Follow-up visit - ₹500

New patient consultation - ₹800

Echo screening - ₹2,500

[Book →](#)**Dr. Kabir Mehta**
Dermatologist

Skin, hair and nail specialist. MD (Dermatology), fellowship in cosmetic dermatology.

Consultation - ₹600

Chemical peel session - ₹1,800

Mole / skin-tag removal - ₹2,200

[Book →](#)*Browse — providers within a category, searchable*

The 3-step booking flow

Book It

DoctorsSalonsTurfsManage booking



Dr. Asha Rao

Cardiologist

15+ years in preventive cardiology. MBBS, MD (Cardiology). Known for unhurried consultations.

WEEKLY HOURS

Sun	Closed
Mon	09:00–13:00, 17:00–20:00
Tue	09:00–13:00, 17:00–20:00
Wed	09:00–13:00
Thu	09:00–13:00, 17:00–20:00
Fri	09:00–13:00
Sat	10:00–14:00

1 Choose a service

Follow-up visit

Review of reports and medication adjustment

₹500

15 min

New patient consultation

Full history, examination and ECG review

₹800

30 min

Echo screening

Echocardiogram with same-visit reading

₹2,500

45 min

BookIt — Appointment Booking System

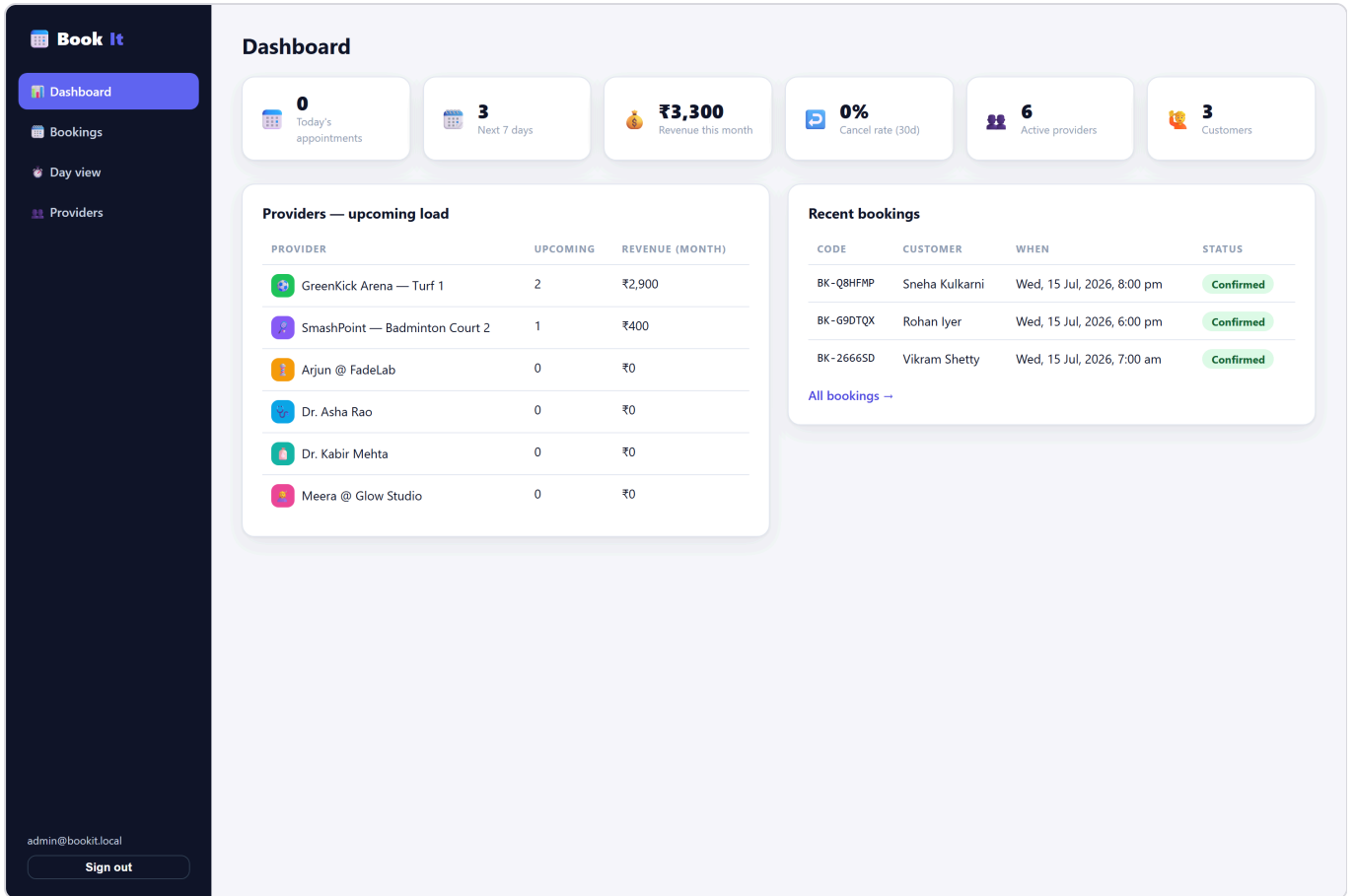
[Admin panel →](#)

Provider detail — profile + weekly hours on the left, the booking flow on the right

- **Step 1 — choose a service** (name, description, price ₹, duration). E.g. Dr. Asha Rao offers Follow-up ₹500 / New patient ₹800 / Echo screening ₹2,500.
- **Step 2 — pick a date & slot** — a 14-day strip (closed days flagged), and time slots grouped into Morning / Afternoon / Evening, **computed live** from the provider's schedule.
- **Step 3 — enter details** (name, email, phone, notes) and confirm. A **409 Conflict** (someone grabbed the slot first) refreshes the grid inline.
- **Confirmation** — a booking code the customer keeps.
- **Manage booking** (/manage) — self-service lookup by **booking code + email**; cancel (auto-releases the slot, sends a cancellation email). Deep-linkable from emails.
- **Email confirmations & cancellations** — HTML emails with the code and a manage link.

2 • The admin panel

Seeded login `admin@bookit.local` / `admin123` (JWT-guarded).



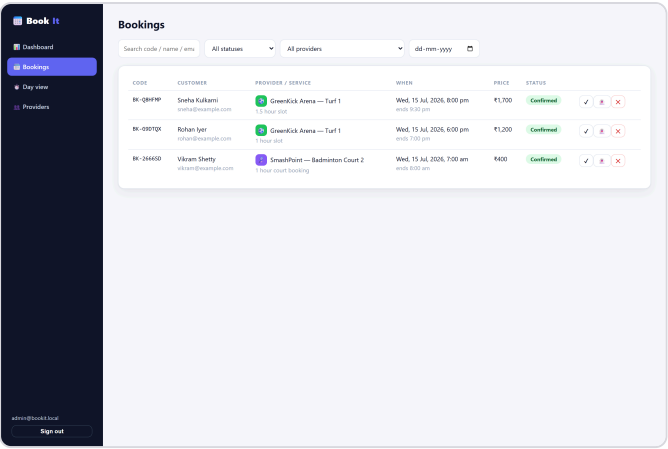
The screenshot shows the 'Book It' Admin Dashboard. On the left is a dark sidebar with navigation links: 'Dashboard' (active), 'Bookings', 'Day view', and 'Providers'. At the bottom of the sidebar, it shows the user 'admin@bookit.local' and a 'Sign out' button. The main content area is titled 'Dashboard' and features six summary tiles: '0 Today's appointments', '3 Next 7 days', '₹3,300 Revenue this month', '0% Cancel rate (30d)', '6 Active providers', and '3 Customers'. Below these tiles are two tables. The 'Providers — upcoming load' table lists providers with their upcoming appointments and monthly revenue. The 'Recent bookings' table lists recent bookings with their codes, customer names, dates, and statuses. A link 'All bookings →' is provided below the recent bookings table.

PROVIDER	UPCOMING	REVENUE (MONTH)
GreenKick Arena — Turf 1	2	₹2,900
SmashPoint — Badminton Court 2	1	₹400
Arjun @ FadeLab	0	₹0
Dr. Asha Rao	0	₹0
Dr. Kabir Mehta	0	₹0
Meera @ Glow Studio	0	₹0

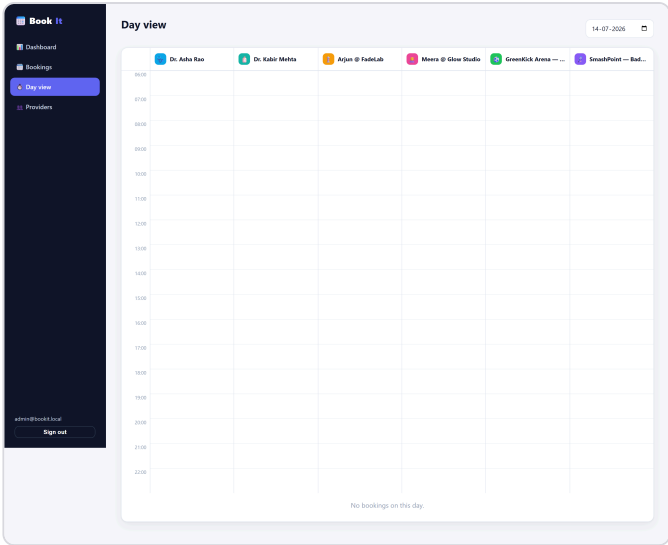
CODE	CUSTOMER	WHEN	STATUS
BK-Q8HFMP	Sneha Kulkarni	Wed, 15 Jul, 2026, 8:00 pm	Confirmed
BK-G9DTQX	Rohan Iyer	Wed, 15 Jul, 2026, 6:00 pm	Confirmed
BK-2666SD	Vikram Shetty	Wed, 15 Jul, 2026, 7:00 am	Confirmed

Admin dashboard — today's load, 7-day pipeline, month revenue, cancel rate, per-provider stats

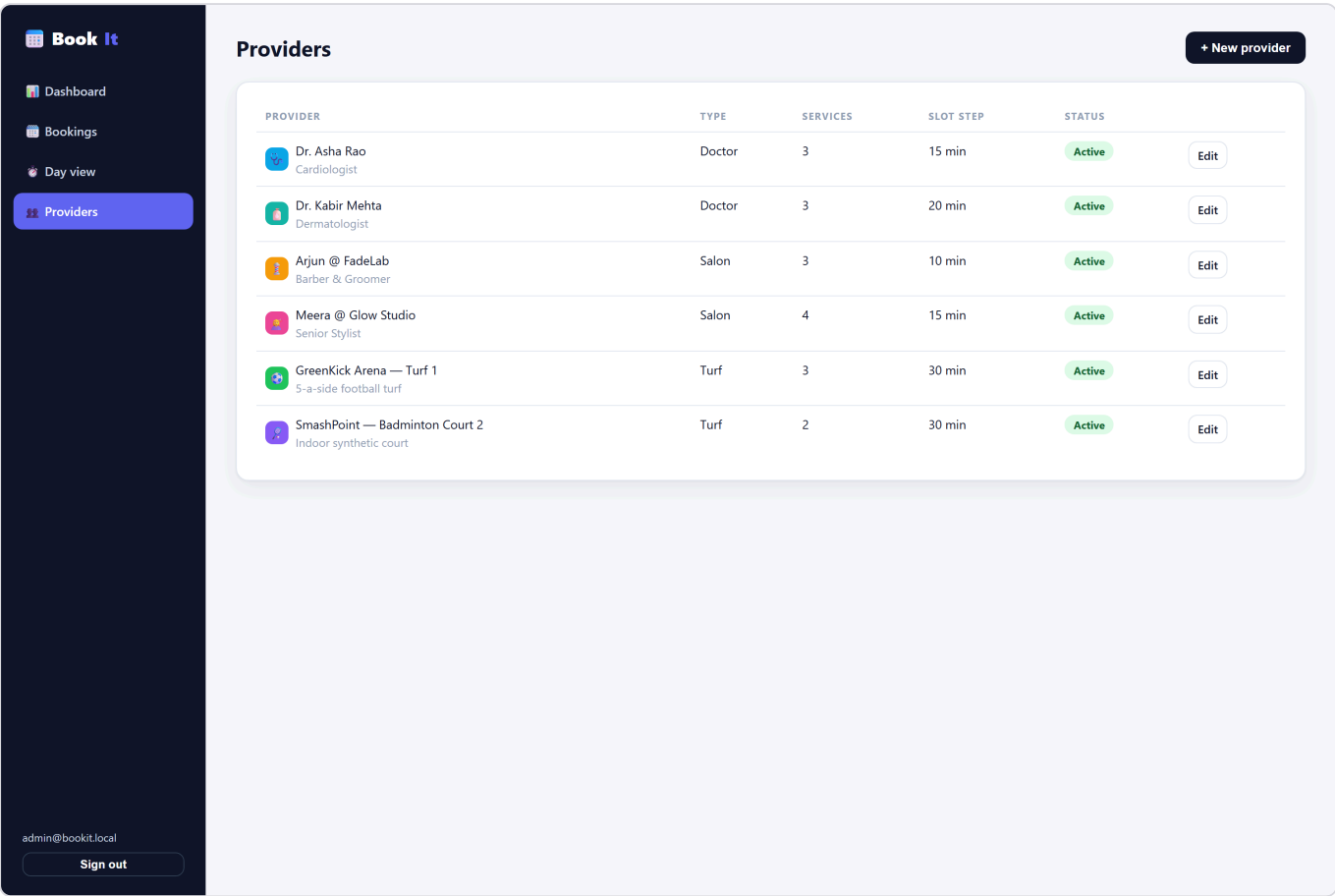
- **Dashboard** — tiles for today's appointments, next-7-days pipeline, month revenue (₹3,300), 30-day cancel rate, active providers (6) and customers (3); a "providers — upcoming load" table and recent bookings.
- **Bookings** — filter by status / provider / date / free-text; actions: complete, no-show, cancel (emails the customer); per-booking event history.
- **Day view** — a visual timeline of all confirmed/completed bookings across providers for a chosen date.
- **Providers** — grouped by business type with service counts; the **provider editor** covers details + booking policies (slot step, lead time, horizon), the weekly schedule (working windows + recurring breaks), one-off time-off, and service CRUD.



Bookings — filters + status actions + audit history



Day view — timeline across providers



Providers — grouped by business type

3 • The flagship: zero double-booking

The whole point of a booking system is that two people can never hold the same slot. BookIt enforces this with **three independent layers**, so even a race between concurrent requests is safe.

Layer 1 — advisory lock. The purchase handler calls `pg_advisory_xact_lock(42, provider_id)`, which **serializes concurrent bookings per provider** — two requests for the same provider queue rather than interleave.

Layer 2 — in-transaction re-validation. Inside the same transaction, the slot is re-checked against the *live* availability engine (not the value the client sent), so a slot taken microseconds earlier is rejected.

Layer 3 — database exclusion constraint. A Postgres GiST `EXCLUSION` constraint (`bookings_no_overlap`, using the `btree_gist` extension and a `tstzrange`) makes an overlapping booking *physically impossible* to insert. The SQLSTATE `23P01` is mapped to **HTTP 409**. This is the last line of defense — even if the app logic had a bug, the database still refuses.

The availability engine

Slots aren't stored — they're **computed**. `services/slots.ts` walks each working window in `slot_step_min` steps and subtracts breaks, one-off time-off, existing bookings, and a per-service buffer, then applies the minimum lead time and the booking horizon. So the grid a customer sees always reflects the true schedule.

4 · Likely interview questions

Q: How do you stop two people booking the same slot?

Three layers: a per-provider advisory transaction lock serializes the critical section, I re-validate the slot inside the transaction against the live engine, and a Postgres GiST exclusion constraint makes an overlap impossible at the database level — its `23P01` error becomes a 409 the client handles by refreshing the grid.

Q: Why raw SQL and `pg` instead of an ORM?

The conflict-prevention relies on Postgres-specific features — advisory locks, `tstzrange`, and GiST exclusion constraints — that are awkward to express through an ORM. Hand-written SQL keeps that logic explicit and auditable.

Q: How are available slots computed?

They're generated on the fly: for the chosen date I walk the provider's working windows in fixed steps and subtract breaks, time-off, existing bookings and a buffer, then apply lead time and horizon. Nothing is precomputed, so it can never drift from the real schedule.

Q: Why no customer accounts?

Customers are identified by a booking code + email, which is enough to confirm, look up, and cancel — it keeps the funnel frictionless. Only admins authenticate (JWT + bcrypt).

Q: How is the audit trail kept?

Every booking writes `booking_events` rows (created, status changes, emails sent, actor), so the admin can see the full history of any booking.

Q: What happens to emails with no SMTP configured?

nodemailer falls back to a JSON transport that writes the rendered HTML to `server/outbox/`, so the flow is demonstrable offline.